The double-slit experiment

Voiceover script · about 63 seconds · 190 words · read at about 3 words per second

Chapter 1 - Bullets and waves (0:00)

Physics, please. What is this quantum physics nonsense? One electron. Two gaps. It went through both. I can't even get through one door without hitting the frame. Fire paintballs at a wall with two slits, and you get two stripes. Makes sense. Nobody's crying. But water waves make many stripes. Where peaks meet, waves grow. Where a peak meets a dip, they cancel. That's an interference pattern.

Chapter 2 - One electron at a time (0:22)

Then scientists fired electrons, one at a time. Each one lands as a single dot, like a tiny bullet. But thousands of dots build the stripes. One electron, alone, interfering with itself. Like using both doors of a shop and meeting yourself at the till.

Chapter 3 - The detector (0:36)

Scientists got suspicious and put a detector at the slits. The pattern vanished. Two boring stripes. You see, the detector has to interact with the electron to record its slit. Once that record exists, the pattern's gone. Camera or coffee cup, it makes no difference. ‘Observer’ doesn't mean a person. So until you measure it, the electron acts like a spread-out wave of possibilities. Measure it, and you get one answer. Electrons are particles, waves, and drama queens.